

Aim:

Write a Python program to classify temperatures in degrees Celsius based on the following conditions:

- If the temperature is less than **-273.15**, print: "Invalid temperature: below absolute zero".
- If the temperature is exactly **-273.15**, print: "Temperature is absolute zero".
- If the temperature is between **-273.15** and **0**, print: "Temperature is below freezing".
- If the temperature is exactly **0**, print: "Temperature is at the freezing point".
- If the temperature is between **0** and **100**, print: "Temperature is in the normal range".
- If the temperature is exactly **100**, print: "Temperature is at the boiling point".
- If the temperature is greater than **100**, print: "Temperature is above the boiling point".

Input Format:

- A single float value representing the temperature in degrees Celsius.

Output Format:

- A message describing the temperature classification as per the conditions.

Sample Input:

Enter temperature in degree Celsius: -273.15

Sample Output:

Temperature is absolute zero

Source Code:

temperature_check.py

```
temp=float(input("Enter temperature in degree Celsius: "))
if(temp<-273.15):
    print("Invalid temperature: below absolute zero")
elif(temp== -273.15):
    print("Temperature is absolute zero")
elif(-273.15<temp<0):
    print("Temperature is below freezing")
elif(temp==0):
    print("Temperature is at the freezing point")
elif(0<temp<100):
    print("Temperature is in the normal range")
elif(temp==100):
    print("Temperature is at the boiling point")
else:
    print("Temperature is above the boiling point")
```

Execution Results - All test cases have succeeded!

Test Case - 1

User Output

Enter temperature in degree Celsius:
-273.15
Temperature is absolute zero

Test Case - 2
User Output
Enter temperature in degree Celsius:
-274
Invalid temperature: below absolute zero

Test Case - 3
User Output
Enter temperature in degree Celsius:
-100
Temperature is below freezing

Test Case - 4
User Output
Enter temperature in degree Celsius:
0.1
Temperature is in the normal range

Test Case - 5
User Output
Enter temperature in degree Celsius:
99.999
Temperature is in the normal range

Test Case - 6
User Output
Enter temperature in degree Celsius:
100
Temperature is at the boiling point

Test Case - 7
User Output
Enter temperature in degree Celsius:
101
Temperature is above the boiling point